PROJECT REPORT

ON

“Arya College ClubSphere: Centralized Club and Event Management Platform”

Submitted in partial fulfillment of the requirement for the award of Degree of

Bachelor of Technology in Computer Science & Engineering

Submitted to:

Rajasthan Technical University, Kota (Raj.)

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PBL Team ID: PBL2627-CSE-A-024

Under the guidance of

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Session: 2026–27

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CERTIFICATE

This is to certify that the project report entitled “Arya College ClubSphere: Centralized Club and Event Management Platform”, submitted by Avantika Mishra (24EARCS032), Avanshi Kothari (24EARCS031), Akansha Sharma (24EARCS012) and Aditi Maheshwari (24EARCS007) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science & Engineering of Rajasthan Technical University, Kota, is a record of the work carried out by the group at Arya College of Engineering & I.T., Jaipur during the session 2026–27, under the guidance and supervision of Er. Ram Babu Buri.

To the best of our knowledge, the work embodied in this report has not been submitted to any other university or institute for the award of any degree or diploma. The project is identified as **PBL Team ID PBL2627-CSE-A-024**.

Date: _____ Place: Jaipur

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We express our sincere gratitude to our project guide, Er. Ram Babu Buri, for his guidance, encouragement and valuable suggestions during the planning and development of Arya College ClubSphere. His emphasis on clear requirements, modular design and systematic verification helped us organise a four-module Java web application around a shared database schema and a common authentication filter.

We sincerely thank Dr. Akhil Pandey, Head of the Department of Computer Science and Engineering, Arya College of Engineering & I.T., for the academic environment and support provided for this Project-Based Learning work. We are also thankful to the faculty members of the department for their cooperation.

We acknowledge the effort of every team member who contributed to planning, implementation, integration and review. Working collaboratively with Git and a shared GitHub repository helped us understand version control, interface contracts between modules, and the need to keep secrets out of source control.

Finally, we thank our families and friends for their encouragement and support throughout the project.

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PBL2627-CSE-A-024

ABSTRACT

Arya College ClubSphere is a web-based club and event management system developed for Arya College of Engineering & I.T., Jaipur. College clubs were previously coordinated through informal chats, posters and spreadsheets. ClubSphere provides one portal for administrators, club heads, club members and students to manage the official club directory, publish events, register (including through QR codes), assign club tasks and receive in-app notifications.

The application is a server-rendered Java web system: Jakarta Servlets and JSP on Apache Tomcat 10.1, JDBC with HikariCP, MySQL 8, BCrypt password hashing and ZXing QR generation. It is packaged as a Maven WAR and can be demonstrated locally, with Docker Compose, or on the hosted demo at https://event-management-uylc.onrender.com/.

This report describes the problem context, requirements, technology stack, layered architecture, authentication and QR registration flows, conceptual data model, module design, security considerations, testing approach and future scope. UML diagrams prepared for the project are included as figures. The report does not claim unmeasured performance results.

Keywords: Java Servlets, JSP, MySQL, BCrypt, QR registration, role-based access control, college clubs.

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CHAPTER 1: INTRODUCTION

1.1 Background of the Project

Arya College of Engineering & I.T. hosts a large number of student societies — technical clubs, cultural groups, NSS, sports and others. Event publicity and membership records are often spread across WhatsApp groups, printed posters and personal spreadsheets. Students miss registrations; club heads lack a single roster; administrators cannot see users, clubs and events in one place. ClubSphere is proposed as a college-specific web portal that centralises these workflows without becoming a full learning-management system.

1.2 About the Institution

The project is developed for Arya College of Engineering & I.T., Jaipur, Rajasthan, affiliated with Rajasthan Technical University, Kota. It is a Project-Based Learning (PBL) assignment of the Department of Computer Science & Engineering for session 2026–27. The official team identifier is **PBL2627-CSE-A-024**.

1.3 Problem Statement

Club discovery, event registration and internal club work become fragmented when each society uses a different informal tool. A central portal can improve discoverability and provide role-based access. At the same time, accounts must be protected, passwords must not be stored in plaintext, and administrative operations must be restricted to authorised roles.

1.4 Objectives of the Project

- To develop a web-based portal for Admin, Club Head, Club Member and Student roles.
- To provide secure login using BCrypt password hashing and HTTP sessions.
- To maintain an official directory of campus clubs and memberships.
- To support event publishing, individual/group registration and QR registration links.
- To provide role-specific dashboards, club tasks, announcements and in-app notifications.
- To persist data in MySQL and package the system as a Tomcat WAR (with Docker as an optional demo path).

1.5 Scope of the Project

In scope: authentication and user administration, club directory and membership, events and QR-based registration, dashboards, tasks, announcements, notifications and simple reports. Out of scope for this release: public self-service signup (administrators create accounts), native mobile applications, payment gateways, email/SMS gateways, and a JSON REST API for a separate single-page application. ClubSphere is not an LMS and does not manage courses or assignments.

1.6 Significance and Expected Benefits

- Students: one place to browse 26 seeded clubs and register for events.
- Club heads: roster, events, QR codes, tasks and announcements for the assigned club.
- Administrators: users, clubs, events, registrations and reports in one portal.
- Development team: practical experience with Java web technologies, JDBC, RBAC and Git collaboration.

1.7 Team Organisation and Working Method

The project is developed by four students. Work is divided by module and coordinated through GitHub repository Avantikamishra862/Arya_Clubsphere. Shared files (schema, authentication filter, navbar/sidebar, pom.xml) are treated as integration contracts.

Member	Enrollment	Module
Avantika Mishra	24EARCS032	Authentication & User Management
Avanshi Kothari	24EARCS031	Club Management
Akansha Sharma	24EARCS012	Event Management & QR Code Registration
Aditi Maheshwari	24EARCS007	Dashboards, Tasks & Notifications

1.8 Organisation of the Report

Chapter 2 introduces the technologies. Chapter 3 analyses requirements. Chapter 4 presents architecture and data design with UML figures. Chapter 5 describes the four modules. Chapter 6 discusses security. Chapter 7 presents testing. Chapter 8 concludes the report.

CHAPTER 2: TECHNOLOGIES AND CONCEPTS

2.1 Java, JSP and Jakarta Servlets

Java 17 is the implementation language. Jakarta Servlets on Tomcat 10.1 handle HTTP. JSP pages render HTML for browsers. This is a server-rendered application: form posts and page forwards, not a React SPA.

2.2 HTML, CSS, JavaScript and Bootstrap

Views use HTML, CSS (style.css with Arya red-and-white identity), JavaScript for local interaction (password visibility, demo autofill) and Bootstrap 5 for layout.

2.3 JDBC, HikariCP and MySQL

MySQL 8 stores users, clubs, events, registrations, tasks, announcements, notifications and activity logs. DAOs use PreparedStatement. HikariCP pools connections. DBUtil reads environment variables before application.properties so secrets are not required in Git.

2.4 BCrypt and HTTP Sessions

Passwords are hashed with BCrypt (cost 10). Authentication uses a server-side HTTP session attribute currentUser, not JWT. Login invalidates any previous session before creating a new one.

2.5 ZXing QR Codes

Each event has a unique qr_token. QRCodeServlet returns a PNG whose payload is APP_BASE_URL/event/register?token=… so a phone camera opens the registration page.

2.6 Maven, Tomcat, Docker and Git

Maven builds aryaClgClubSphere.war. Tomcat 10.1 hosts it. A multi-stage Dockerfile copies the WAR as ROOT.war and binds to $PORT for hosts such as Render. docker-compose.yml starts MySQL and the app for a one-command demo. GitHub is used for collaboration.

2.7 Technology Stack Summary

Layer	Technology	Purpose
Frontend	JSP, HTML, CSS, JS, Bootstrap 5	Pages and forms
Backend	Java 17, Jakarta Servlets	Request handling and business rules
Security	BCrypt, AuthenticationFilter, HttpOnly session	Credentials and RBAC
Database	MySQL 8, JDBC, HikariCP	Persistent relational data
QR	ZXing	Event registration QR PNG
Build / run	Maven, Tomcat 10.1, Docker	WAR, container, hosted demo
Version control	Git and GitHub	Team collaboration

CHAPTER 3: REQUIREMENT ANALYSIS

3.1 Existing System and Its Limitations

Informal tools do not encode Arya's club hierarchy, capacity limits, QR tokens or role dashboards. Google Forms can collect registrations but cannot enforce club-head ownership or generate a check-in QR bound to a ClubSphere event record.

3.2 Proposed System

ClubSphere is a browser portal with login, role-aware pages, a public club/event catalogue, QR registration, tasks and notifications. Data is stored in clubsphere_db. Demo accounts are seeded for evaluation.

3.3 User Classes and Roles

Role	Primary responsibility	Access principle
ADMIN	Users, clubs, system-wide events and reports	Only /admin/* plus public pages
CLUB_HEAD	Members, events, QR, tasks, announcements for one club	/club-head/* for the assigned club
CLUB_MEMBER	Assigned tasks, club events, announcements	/club-member/*
STUDENT	Browse clubs/events, register, profile	/student/* and public catalogue

3.4 Functional Requirements

- FR-1: Authenticate with username or email and password.
- FR-2: Store passwords as BCrypt hashes.
- FR-3: Redirect each role to its dashboard after login.
- FR-4: Protect /admin, /club-head, /club-member and /student according to role.
- FR-5: Allow admins to create users, change roles and toggle status (without self-lockout).
- FR-6: Provide public club search/view and admin/club-head membership management.
- FR-7: Allow club heads to create events with a unique QR token.
- FR-8: Allow students to register by eventId or QR token, once per event.
- FR-9: Support group registrations with participant rows.
- FR-10: Provide tasks, announcements, notifications and reports.

3.5 Non-Functional Requirements

Category	Requirement
Security	BCrypt, session fixation handling, RBAC filter, parameterized SQL, env-based secrets
Usability	Clear forms, demo autofill, role dashboards, Bootstrap layout
Maintainability	Servlet → Service → DAO layering
Deployability	Documented WAR, Docker Compose and Render PORT binding
Testability	JUnit tests for PasswordUtil and QRCodeUtil without a live database

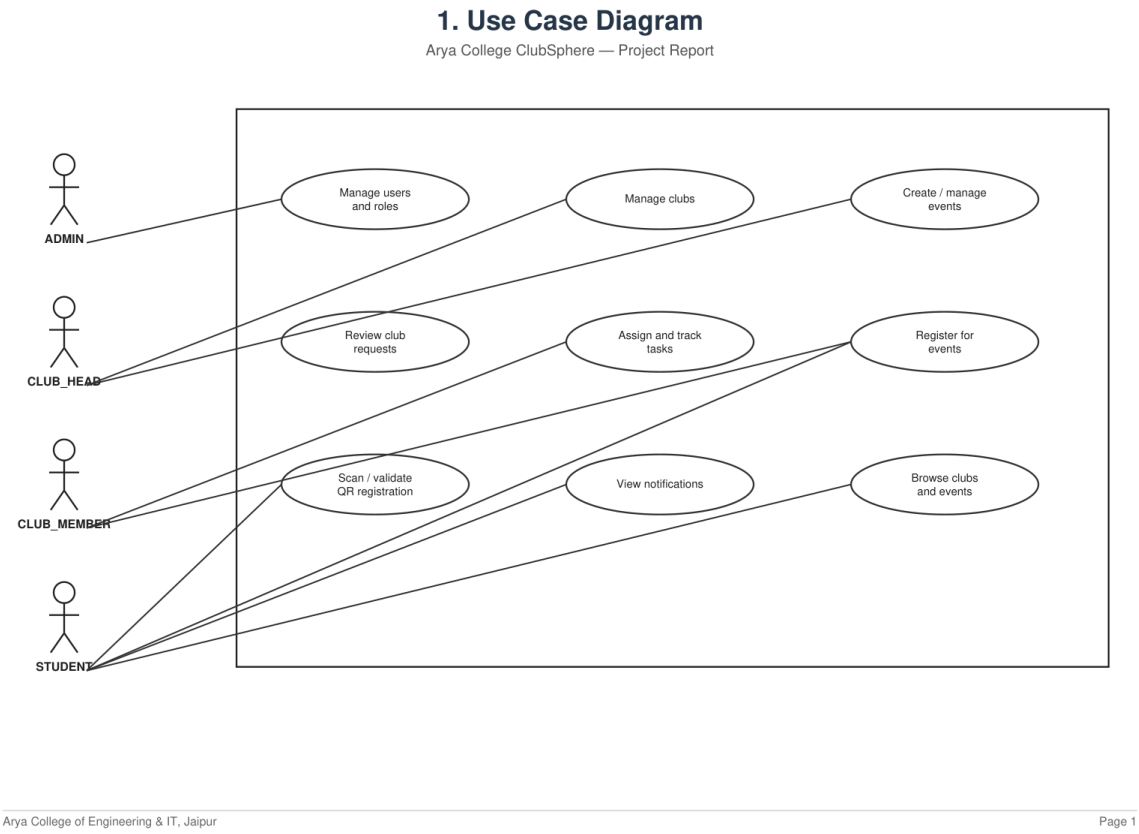
3.6 Design Constraints

The project is an academic PBL with a four-member team. JDK 17 and Tomcat 10.1 (Jakarta namespace) are required. Production passwords must not be committed. QR links depend on a correct APP_BASE_URL for the deployed context path.

3.7 Success Criteria

Success is a clean Maven package, a working login for four demo roles, protected URLs, club and event pages, QR PNG generation, and seeded MySQL data. Quantitative performance claims are not made.

Figure 3.1 — UML Use Case Diagram



Supports Section 3.3 by showing Admin, Club Head, Club Member and Student interactions with ClubSphere.

CHAPTER 4: SYSTEM DESIGN

4.1 Architectural Style

ClubSphere follows a classic three-tier Java web architecture. The browser sends HTTP requests. AuthenticationFilter enforces public whitelist and RBAC. Servlets call services; services call DAOs; DAOs talk to MySQL through HikariCP.

4.2 Layered Architecture

Layer	Responsibilities
Presentation	JSP pages, Bootstrap, navbar/sidebar, forms
Filter	AuthenticationFilter: session check and role prefixes
Controller	Jakarta Servlets mapped with @WebServlet
Service	Login, membership, capacity, notifications
Persistence	DAO + PreparedStatement + HikariCP
Database	MySQL 8 schema clubsphere_db

4.3 Authentication Request Flow

The user posts username/email and password to /login. LoginServlet calls UserService, which loads the user and verifies BCrypt. On success the previous session is invalidated, a new session stores currentUser and unreadCount, an activity log is written, and the user is redirected to the role dashboard. The optional redirect parameter is accepted only for internal paths.

4.4 Database Design

Normative SQL is database/schema.sql. Seed data is database/seed.sql (demo password password123). Core tables: users, clubs, club_members, events, registrations, event_participants, tasks, announcements, notifications, activity_logs.

4.5 Conceptual Entities

Entity	Example attributes	Purpose
User	id, username, email, password_hash, role, status	Identity and login
Club	club_code, name, head_user_id, category, status	Official society
ClubMember	club_id, user_id, member_role	Roster
Event	title, qr_token, capacity, status, venue	Campus event
Registration	registration_number, event_id, user_id	Signup record
Task / Notification	assignee, type, is_read	Club work and inbox

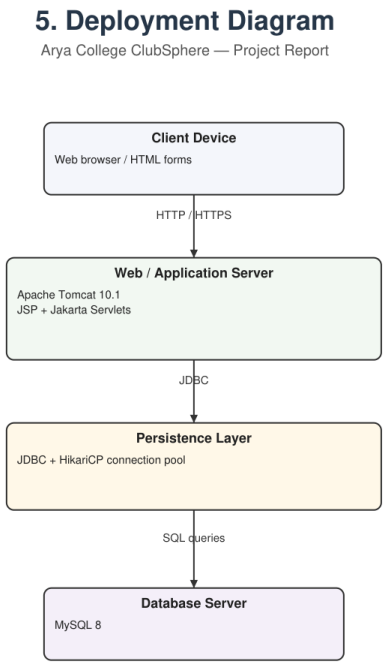
4.6 Request Map (in place of a REST API catalogue)

ClubSphere does not expose a public JSON API. Representative URLs include /login, /clubs, /events, /event/register, /qr/{token}, /admin/*, /club-head/*, /club-member/*, /student/* and /notifications/*.

4.7 Role-Based Access Control

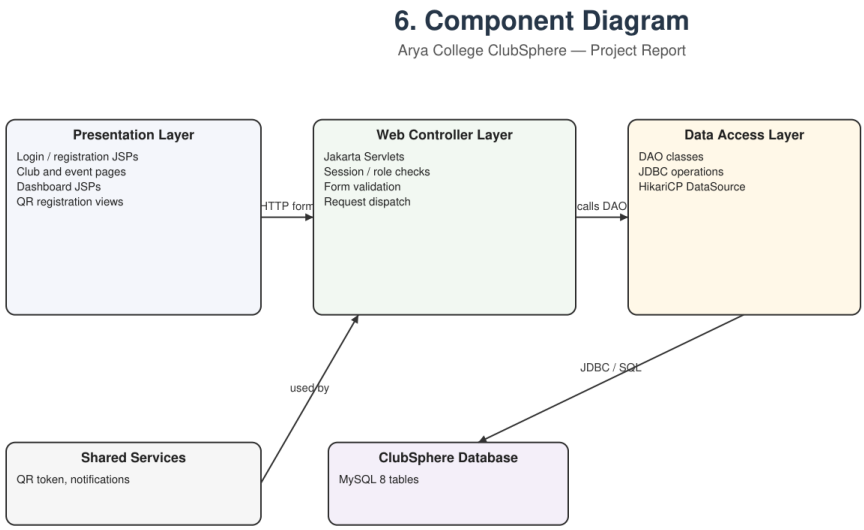
AuthenticationFilter is the security boundary. The UI may hide links, but URL typing is still blocked with a login redirect or HTTP 403. Club-head screens additionally require an assigned club (head_user_id).

Figure 4.1 — UML Deployment Diagram



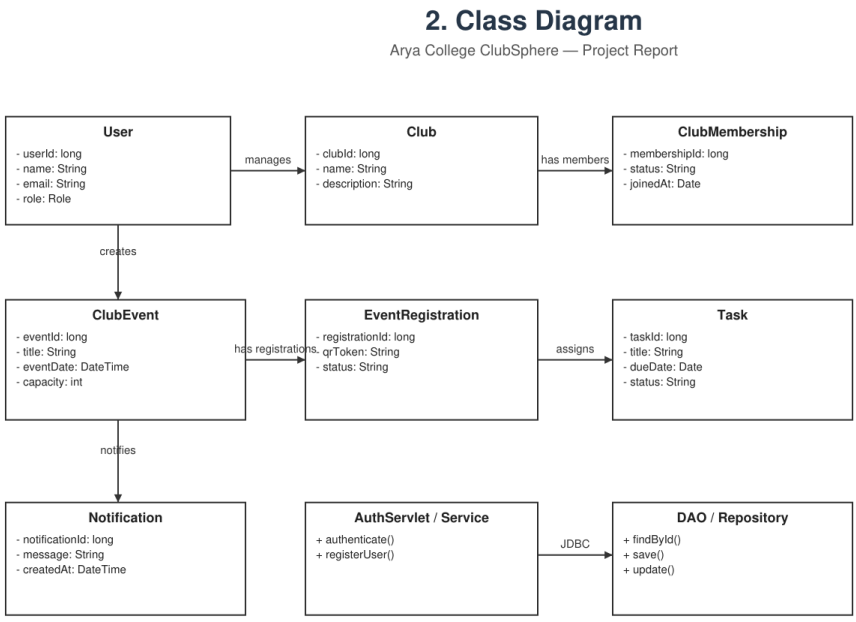
Client browser, Apache Tomcat 10.1 and MySQL 8 as described in Sections 4.1–4.2.

Figure 4.2 — UML Component Diagram



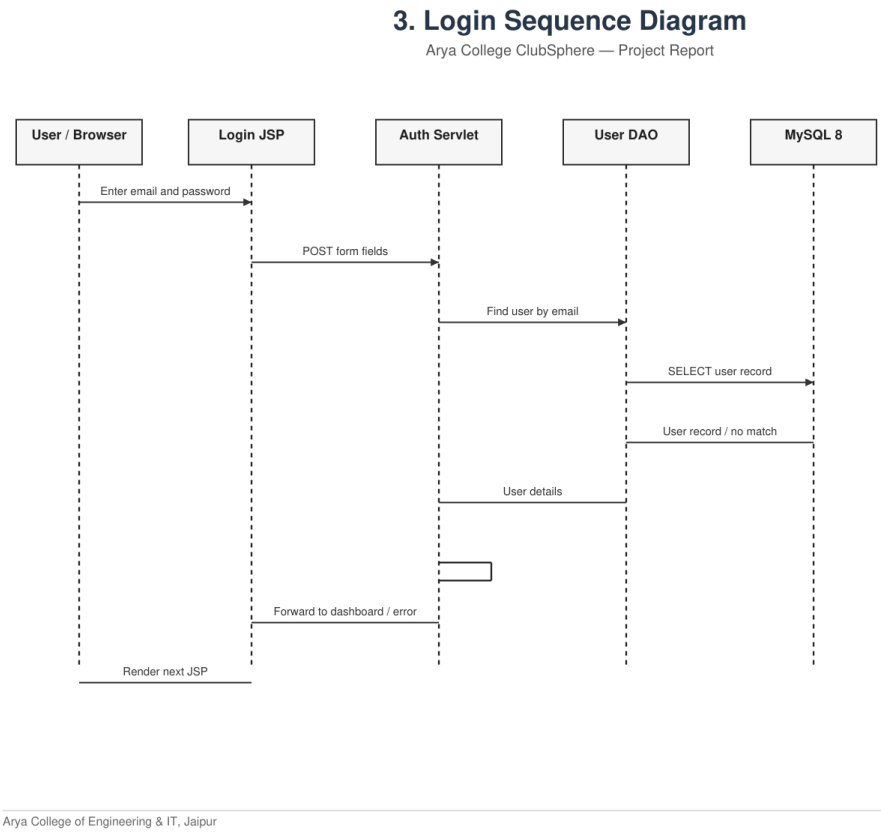
Presentation, servlet, service/DAO and MySQL components of ClubSphere.

Figure 4.3 — UML Class Diagram



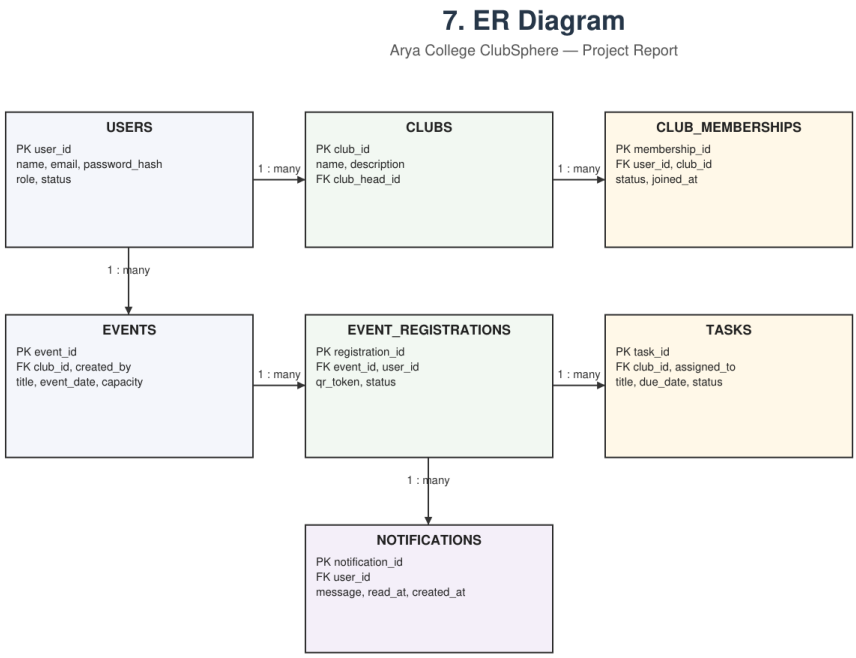
Conceptual domain model; class names in code include User, Club, Event, Registration, Task and Notification.

Figure 4.4 — UML Login Sequence Diagram



Visualises the authentication request flow in Section 4.3.

Figure 4.5 — ER Diagram



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Conceptual data model referenced in Sections 4.4–4.5. Physical names are in database/schema.sql.

CHAPTER 5: MODULE DESIGN AND IMPLEMENTATION APPROACH

5.1 Authentication & User Management (Avantika Mishra)

LoginServlet, LogoutServlet, AuthenticationFilter, UserService, UserDAO and PasswordUtil implement identity. Admins create users, change roles and toggle ACTIVE/INACTIVE/SUSPENDED. An administrator cannot deactivate the currently logged-in admin account.

5.2 Club Management (Avanshi Kothari)

ClubServlet serves public explore/view. AdminServlet manages club CRUD and status. ClubHeadServlet maintains membership for the assigned club. Twenty-six clubs are seeded (coding, robotics, cultural, NSS, sports and others).

5.3 Event Management & QR (Akansha Sharma)

Club heads create events with schedule, venue, capacity and participation type. A unique qr_token is stored. QRCodeServlet emits a ZXing PNG. RegistrationServlet accepts eventId or token, prevents duplicates, and stores group participants when required.

5.4 Dashboards, Tasks & Notifications (Aditi Maheshwari)

Each role has a dashboard JSP. Club heads assign tasks and publish announcements. Members update task status. NotificationServlet lists the inbox and marks items read. ReportService supplies counts for admin and club-head reports.

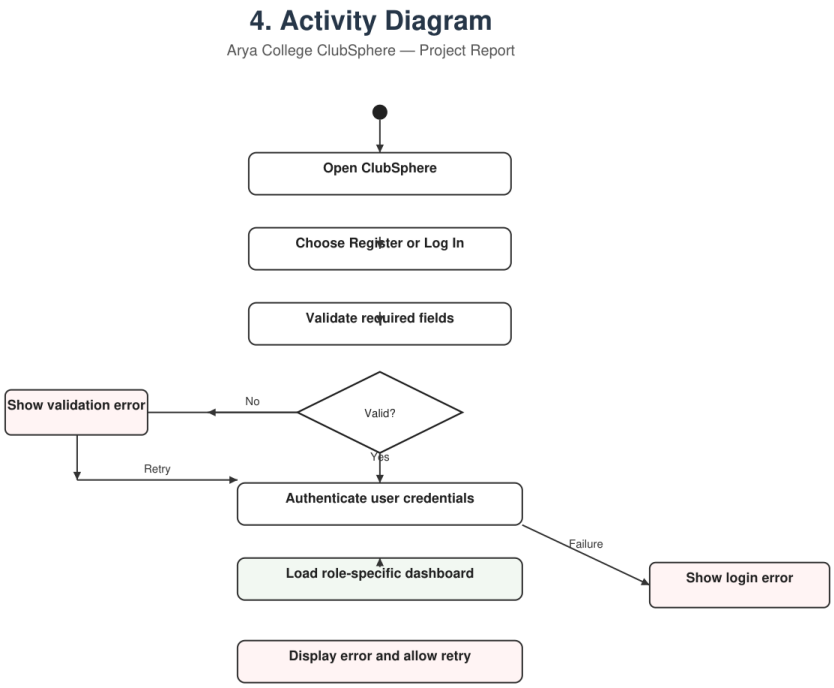
5.5 Configuration and Deployment

Environment variables DB_URL, DB_USERNAME, DB_PASSWORD and APP_BASE_URL override application.properties. A committed password was replaced with a placeholder. Docker Compose provides MySQL plus the application for demonstration.

5.6 Suggested Implementation Sequence

Schema and seed → authentication filter and login → clubs → events/QR → dashboards, tasks and notifications → documentation and Docker.

Figure 5.1 — UML Activity Diagram



Registration and login activity supporting the identity module (Chapter 5.1).

CHAPTER 6: SECURITY ARCHITECTURE

6.1 Security Philosophy

ClubSphere stores credentials and personal student data. Authorisation is enforced in AuthenticationFilter and in servlets, not only by hiding menu items.

6.2 Password Security

Passwords are BCrypt hashes. Demo seed accounts use a documented evaluation password. Real deployment passwords must be unique and uncommitted.

6.3 Session Handling

Sessions are server-side, HttpOnly, 60-minute timeout. Login performs session-fixation mitigation. Logout invalidates the session.

6.4 Open Redirect and RBAC

The login redirect parameter must be an internal path. Role prefixes are enforced as described in Chapter 3.

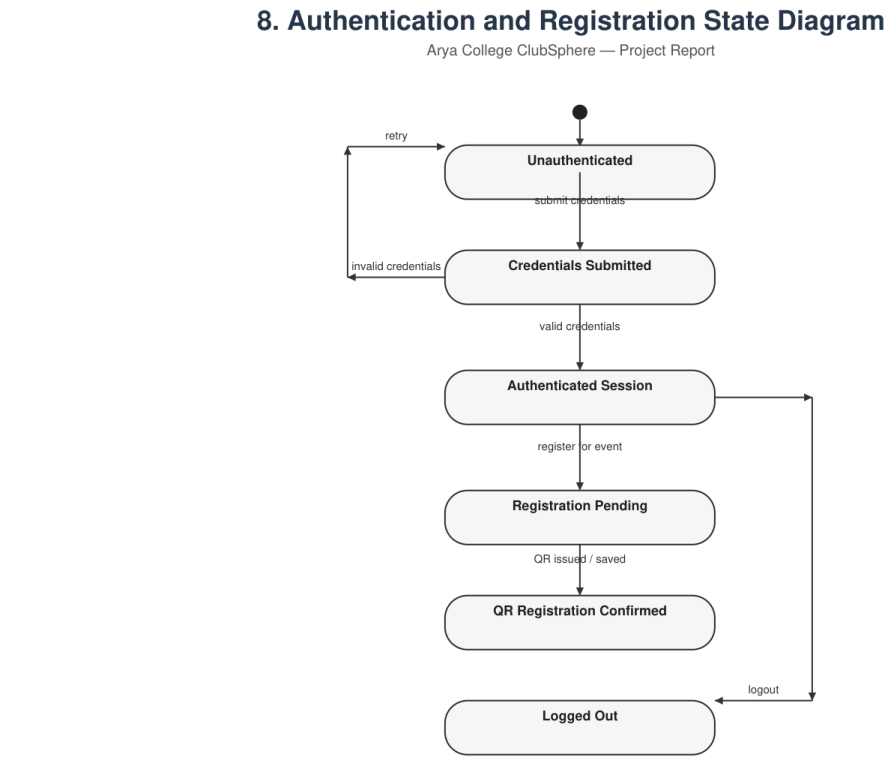
6.5 Secrets, SQL and Errors

DAOs use parameterized SQL. Error pages should not print stack traces to end users. Database passwords belong in environment variables.

6.6 Threats and Mitigations

Threat	Mitigation
Stolen plaintext password	BCrypt hashing; no passwords in Git
Session fixation	Invalidate old session on login
URL tampering to /admin	AuthenticationFilter 403
SQL injection	PreparedStatement in DAOs
Open redirect after login	Internal-path check
QR pointing at wrong host	Configure APP_BASE_URL per environment

Figure 6.1 — Authentication and Registration State Diagram



Login, authenticated session, QR registration and logout states.

CHAPTER 7: DEVELOPMENT PLAN, TESTING AND EVALUATION

7.1 Development Roadmap

Phase	Deliverable
1	Maven WAR, Tomcat, MySQL schema and seed
2	Login, logout, RBAC filter, admin users
3	Club directory and membership
4	Events, registrations, QR PNG
5	Dashboards, tasks, notifications, reports
6	Documentation, env template, Docker Compose, hosted demo

7.2 Testing Strategy

Unit tests cover password hashing and QR PNG generation without MySQL. Manual tests use the four demo roles. Security smoke tests attempt unauthenticated /admin access and a student opening /club-head.

7.3 Representative Test Cases

ID	Scenario	Expected result
T1	Login admin / password123	Admin dashboard
T2	Invalid password	Error on login.jsp
T3	Student GET /admin/dashboard	403 or login redirect
T4	mvn test	PasswordUtilTest and QRCodeUtilTest pass
T5	Register twice for same event	Duplicate rejected
T6	GET /qr/{token}	PNG image
T7	Club head without assigned club	Error page, not another club’s data

7.4 Current Status

A working portal with seeded clubs and events, a public landing page, role dashboards and a hosted demo is available. Automated UI tests and DAO integration tests remain future work. Unit tests for hashing and QR generation passed during documentation work.

7.5 Risks

Risk	Response
Wrong APP_BASE_URL breaks QR	Document WAR vs Docker context paths
Secrets in Git	Placeholder properties and .env.example
Merge conflicts on filter/schema	Coordinate shared files; review before merge

CHAPTER 8: CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

Arya College ClubSphere (PBL2627-CSE-A-024) is a centralized club and event portal for Arya College of Engineering & I.T. It combines JSP/Servlets, MySQL, BCrypt sessions and QR registration into one WAR. The four modules integrate through a shared schema and AuthenticationFilter.

8.2 Limitations

- Administrators create accounts; there is no public self-registration.
- Notifications are in-app only (no email/SMS).
- QR encodes the registration URL; door-scan attendance can be extended later.
- Automated tests currently cover utilities, not full DAO/UI journeys.

8.3 Future Scope

- Public student registration with college-email validation.
- Email reminders for deadlines and event changes.
- Attendance marking by scanning a second check-in QR.
- Richer reports and export (CSV) for club heads.
- Additional integration tests and CI.

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